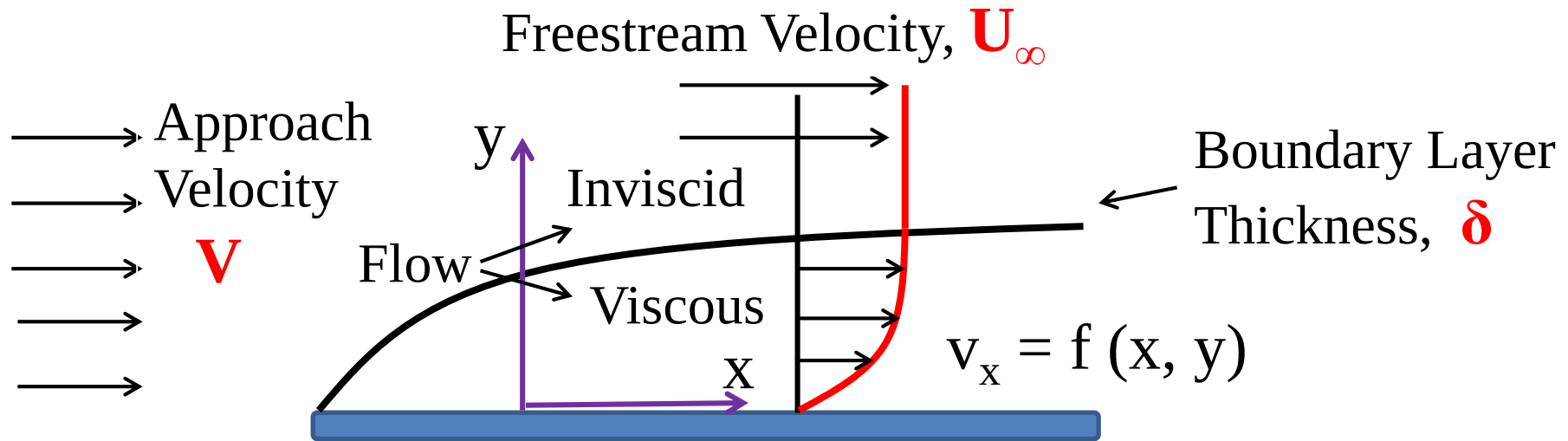


External Incompressible Viscous Flow – Boundary Layer



Flow over a flat plate

$v_x = f(x, y)$ Viscous 2D flow inside BL

$v_x = U_\infty$ Inviscid flow outside BL

$v_x = 0.99U_\infty$
at $y = \delta$

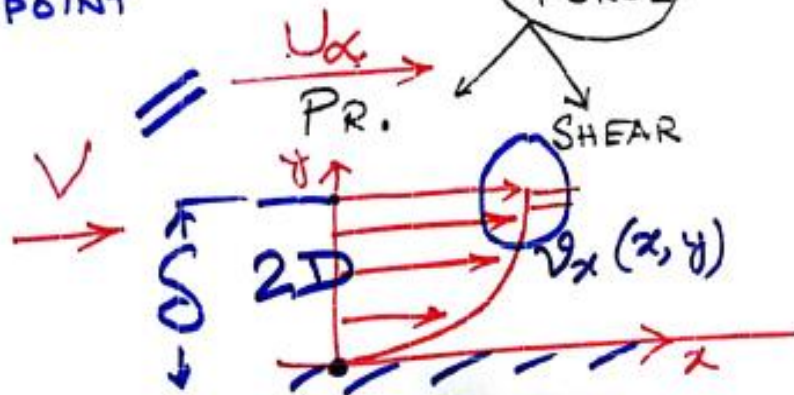
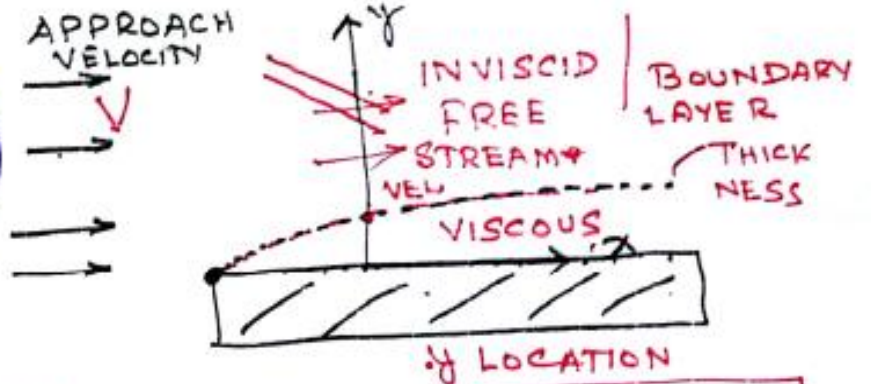
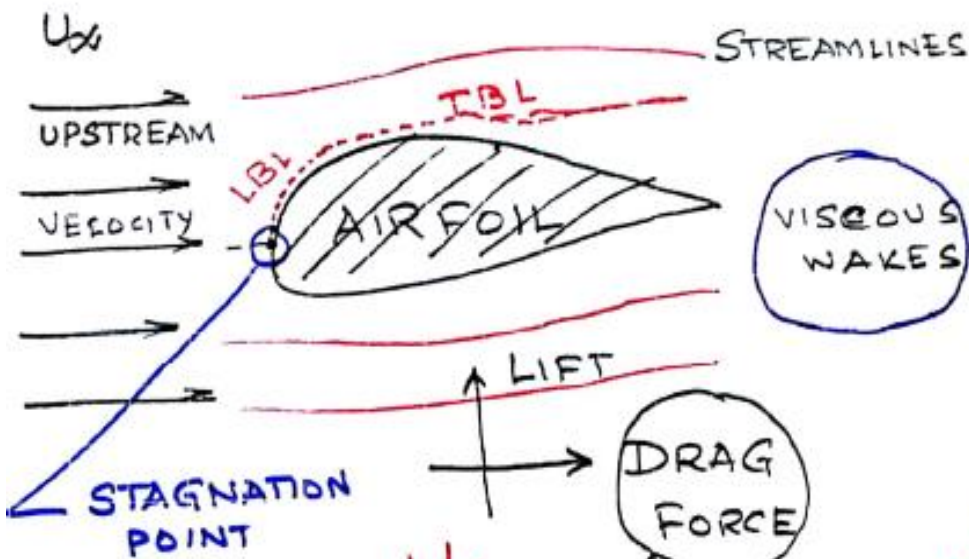
δ - boundary layer thickness

Boundary Layers

EXTERNAL INCOMPRESSIBLE VISCOUS FLOW - BOUNDARY LAYER CONCEPTS

C. GET
IIT KGP

FLUID MECHANICS
Fox & McDonald



$$U_\infty$$

$$V = U_\infty$$

$$v_x \approx 0.99V$$

$$\delta \approx 0.99U_\infty$$

$$v_x \approx 0.99U_\infty$$

y at which

Flow Inside the Boundary Layer

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)$$

Boundary Layer Approximations

$$v_x \gg v_y \quad \frac{\partial v_x}{\partial y} \gg \frac{\partial v_x}{\partial x} \quad \frac{\partial^2 v_x}{\partial y^2} \gg \frac{\partial^2 v_x}{\partial x^2}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \frac{\partial^2 v_x}{\partial y^2}$$

Exact Method – Blasius – Numerical Solution

Approximate Method – Momentum Integral Equation

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \frac{\partial^2 v_x}{\partial y^2}$$

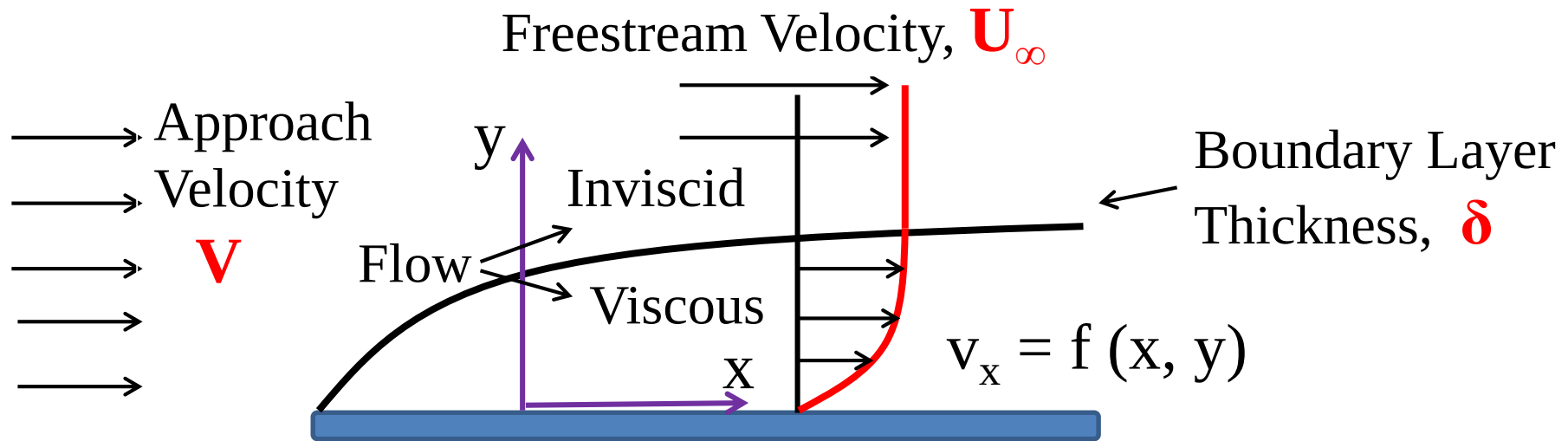
Boundary Conditions

$$\text{At } y = 0; \quad v_x, v_y = 0$$

$$\text{At } y = \delta; \quad v_x = U_\infty$$

$$\text{At } x = 0; \quad v_x = V$$

Boundary Layer Thicknesses



Flow over a flat plate

Boundary Layer Thickness, δ

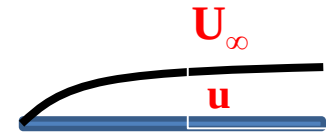
$$y \text{ at which } v_x = 0.99 U_\infty$$

Also known as the disturbance thickness

Displacement Thickness

Amount of reduction in mass flow rate in viscous flow inside the boundary layer

$$= \int_0^{\infty} \rho (U_{\infty} - u) dy$$



For an inviscid flow the same reduction can be achieved by moving the plate up by a distance δ^*



$$\rho U_{\infty} \delta^* = \int_0^{\infty} \rho (U_{\infty} - u) dy$$

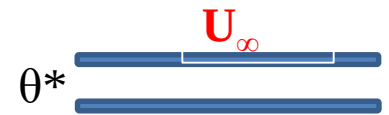
$$\delta^* = \int_0^{\infty} \left(1 - \frac{u}{U_{\infty}} \right) dy \approx \int_0^{\delta} \left(1 - \frac{u}{U_{\infty}} \right) dy$$

Momentum Thickness

Reduction in momentum for the fluid that actually flows in the boundary layer

$$= \int_0^{\infty} \rho u (U_{\infty} - u) dy$$

For an inviscid flow the same momentum reduction can be achieved by moving the plate up by a distance θ (Reduction = $\rho U_{\infty} \theta^* U_{\infty}$)



$$\rho U_{\infty}^2 \theta = \int_0^{\infty} \rho u (U_{\infty} - u) dy$$

$$\theta = \int_0^{\infty} \frac{u}{U_{\infty}} \left(1 - \frac{u}{U_{\infty}} \right) dy \approx \int_0^{\delta} \frac{u}{U_{\infty}} \left(1 - \frac{u}{U_{\infty}} \right) dy$$

Blasius Solution

BLASIUS

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$$\frac{v_x}{U} = f(\eta) \quad \eta \sim \frac{y\sqrt{U}}{\delta(x)}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad (2)$$

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (1)$$

$$v_x = \frac{\partial \psi}{\partial y}, \quad v_y = -\frac{\partial \psi}{\partial x}$$

ψ = STREAM
FUNCTION

$$\frac{\partial \psi}{\partial x \partial y} - \frac{\partial \psi}{\partial y \partial x} = 0$$

ψ EXACT
DIFFERENTIAL

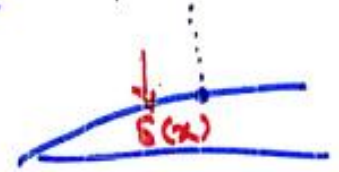
Blasius Solution – contd.

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$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

NEAR THE B.L.

$$v_x \sim U, \quad y \sim \delta, \quad \frac{\partial v_x}{\partial y} \approx 0$$



$$U \frac{U}{x} \sim \nu \frac{U}{\delta^2}$$

$$\delta^2 \sim \frac{\nu x}{U}$$

$$\rightarrow \boxed{\delta \sim \sqrt{\frac{\nu x}{U}}}$$

$f \quad \eta$

$$\eta = \frac{y}{\delta} = \sqrt{\frac{U}{\nu x}} y$$

Blasius Solution – contd.

$$\rightarrow v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

i) INTRODUCING A STREAM FN ψ .

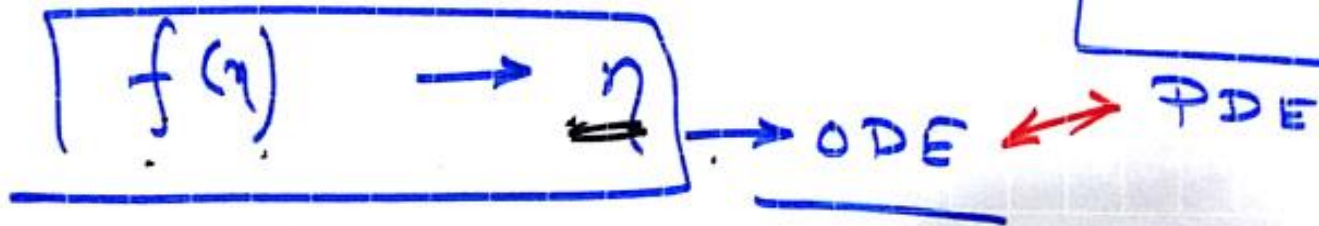
ii) BY INVOKING THE METHOD OF COMBINATION OF VARIABLES

$$\eta = y \sqrt{\frac{U}{\nu x}}$$

DIMENSIONLESS STR. FN

$$f(\eta) = \frac{\psi}{\sqrt{\nu x U}}$$

$$\begin{array}{l} v_x, v_y \\ \rightarrow x, y \end{array}$$



Blasius Solution – contd.

Fox &
McDONALD
FLUID
MECH.

$$v_x = \frac{\partial \psi}{\partial y}$$

$$= \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial y}$$

$$= \sqrt{\nu x U} \left(\frac{df}{d\eta} \right) \sqrt{\frac{U}{\nu x}}$$

$$= U \frac{df}{d\eta}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

$$\psi = \sqrt{\nu x U} f(\eta)$$

$$\eta = y \sqrt{\frac{U}{\nu x}}$$

$$f = \frac{\psi}{\sqrt{\nu x U}}$$

$$v_x = U \frac{df}{d\eta}$$

$$v_y = -\frac{\partial \psi}{\partial x} = -\frac{\partial}{\partial x} [f \cdot \sqrt{\nu x U}]$$

$$= - \left[\sqrt{\nu x U} \cdot \frac{\partial f}{\partial x} + \frac{1}{2} \sqrt{\frac{\nu U}{x}} f \right]$$

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Blasius Solution – contd.

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$$v_y = - \left[\sqrt{\nu x U} \cdot \frac{df}{d\eta} \cdot \frac{\partial \eta}{\partial x} + \frac{1}{2} \sqrt{\frac{\nu U}{x}} f \right]$$

$\Rightarrow$

$$v_y = \frac{1}{2} \sqrt{\frac{\nu U}{x}} \left[\eta \frac{df}{d\eta} - f \right]$$

$$v_x = U \frac{df}{d\eta}$$

$$v_x \left(\frac{\partial v_x}{\partial x} \right) + v_y \left(\frac{\partial v_x}{\partial y} \right) = \nu \left(\frac{\partial^2 v_x}{\partial y^2} \right)$$

$$2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0$$

ODE

$$2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0$$

$$\eta = y \sqrt{\frac{U}{\nu x}} \rightarrow \begin{array}{l} v_x, v_y = 0 \quad \eta = 0 \\ v_x = U \quad \eta \rightarrow \infty \end{array}$$

Blasius Solution – contd.

$$2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0$$

BC. $\eta = 0$

$\eta = \infty$

$f = \frac{df}{d\eta} = 0$
 $f' = 1$

$\eta = y \sqrt{\frac{U}{2x}}$
 $v_x = U \frac{df}{d\eta}$
 $v_y = \frac{1}{2} \sqrt{\frac{2U}{x}} \left[\eta \frac{df}{d\eta} - f \right]$

NUM. SOLN
 HOWARTH

η	f	f'	f''
0	0	0	0.332
5.0	3.28329	0.99155	0.01591
→ 8.0	6.27923	~ 1.0	0.00001
→ 8.4	6.67923	~ 1.0	0.000001

$\eta - f$

Table 9.1 The function $f(\eta)$ for the laminar boundary layer along a flat plate at zero incidence. (After L. Howarth [5].)

$\eta = y \sqrt{\frac{U_\infty}{\nu x}}$	f	$f' = \frac{u}{U_\infty}$	f''
0	0	0	0.33206
0.4	0.02656	0.13277	0.33147
1.0	0.16557	0.32979	0.32301
1.4	0.32298	0.45627	0.30787
2.0	0.65003	0.62977	0.26675
2.4	0.92230	0.72899	0.22809
3.0	1.39682	0.84605	0.16136
3.4	1.74696	0.90177	0.11788
4.0	2.30576	0.95552	0.06424
4.4	2.69238	0.97587	0.03897
5.0	3.28329	0.99155	0.01591
5.4	3.68094	0.99616	0.00793
6.0	4.27964	0.99898	0.00240
6.4	4.67938	0.99961	0.00098
7.0	5.27926	0.99992	0.00022
7.4	5.67924	0.99998	0.00007
8.0	6.27923	1.00000	0.00001
8.4	6.67923	1.00000	0.00000

Blasius Solution – contd.

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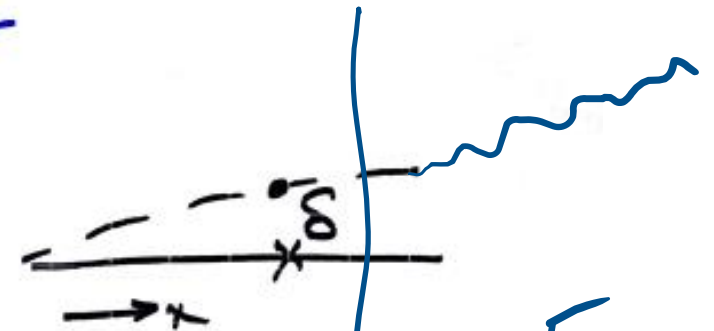
$$\eta = y \sqrt{\frac{U}{2\nu x}}$$

$\eta = 5.0$ EDGE OF THE B.L. $f' = 0.991$
 $\frac{v_x}{U} = 0.99$

$$5.0 = \delta \sqrt{\frac{U}{2\nu x}} \rightarrow$$

$$\delta = \frac{5.0}{\sqrt{U/2\nu x}} = \frac{5.0 x}{\sqrt{Re_x}}$$

$$\delta = \frac{5.0 x}{\sqrt{Re_x}}$$



$$Re_{x,c} = 5 \times 10^5$$

Blasius Solution – contd.

$$\tau_w = \mu \frac{\partial v_x}{\partial y} \Big|_{y=0}$$

$$= \mu \frac{\partial}{\partial y} \frac{\partial \psi}{\partial y} \Big|_{y=0}.$$

$$= \mu \frac{\partial}{\partial y} U \frac{df}{d\eta} \Big|_{\eta=0}.$$

$$= \mu U \frac{d^2 f}{d\eta^2} \cdot \frac{\partial \eta}{\partial y} \Big|_{\eta=0}$$

$$\begin{cases} \eta = y \sqrt{\frac{U}{\nu x}} \\ \frac{\psi}{\sqrt{\nu x U}} = f, \end{cases}$$

0.332

$$\Rightarrow \tau_w = \mu U \sqrt{\frac{U}{\nu x}} \frac{d^2 f}{d\eta^2} \Big|_{\eta=0}.$$

$$\eta = 0 \quad f'' = 0.332$$

$$\tau_w = 0.332 U \sqrt{\rho \mu U / x} = \frac{0.332 \rho U^2}{\sqrt{Re_x}}$$

Blasius Solution – contd.

$$\tau_w = \frac{0.332 \rho U^2}{\sqrt{Re_x}}$$

• SHEAR STRESS COEFF. C_f

$$C_f \equiv \frac{\tau_w}{\frac{1}{2} \rho U^2} = \frac{0.664}{\sqrt{Re_x}}$$
$$\delta = \frac{5.0x}{\sqrt{Re_x}}$$

Average Shear Stress $\bar{\tau}_w = \frac{1}{L} \int_0^L C_f \frac{1}{2} \rho U^2 dx$

Use the numerical results of Howarth to evaluate the following quantities for laminar boundary layer flow on a flat plate

Table 9.1 The function $f(\eta)$ for the laminar boundary layer along a flat plate at zero incidence. (After L. Howarth [5].)

$\eta = y \sqrt{\frac{U_\infty}{\nu x}}$	f	$f' = \frac{u}{U_\infty}$	f''
0	0	0	0.33206
0.4	0.02656	0.13277	0.33147
1.0	0.16557	0.32979	0.32301
1.4	0.32298	0.45627	0.30787
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6.0	4.27964	0.99898	0.00240
6.4	4.67938	0.99961	0.00098
7.0	5.27926	0.99992	0.00022
7.4	5.67924	0.99998	0.00007
8.0	6.27923	1.00000	0.00001
8.4	6.67923	1.00000	0.00000

(a) δ^*/δ evaluate for $\eta = 5$ and $\eta \rightarrow \infty$

(b) v_y / U at the edge of the BL

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U_\infty}\right) dy \approx \int_0^\delta \left(1 - \frac{u}{U_\infty}\right) dy$$

(a) δ^*/δ evaluate for $\eta = 5$ and $\eta \rightarrow \infty$

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U_\infty}\right) dy \approx \int_0^\delta \left(1 - \frac{u}{U_\infty}\right) dy$$

$$\eta = y \sqrt{\frac{U}{\nu x}}$$

$$dy = d\eta \sqrt{\frac{\nu x}{U}}$$

$$\delta^* = \int_0^\eta \left(1 - \frac{u}{U}\right) \sqrt{\frac{\nu x}{U}} d\eta$$

$$\delta^* = \sqrt{\frac{\nu x}{U}} \int_0^\eta \left(1 - \frac{df}{d\eta}\right) d\eta$$

$$\delta^* = \delta_{1/5}(\checkmark)$$

$\frac{\delta^*}{\delta}$
 $\frac{u}{U_\infty}$
 $\frac{\nu}{U_\infty}$
 $\frac{x}{U_\infty}$

$$\delta^* = \frac{1}{5} \left[\int_0^{\eta} (1 - f') d\eta \right]$$

$$\checkmark \frac{\delta^*}{\delta} = \frac{1}{5} \left[\eta - f \right]_0^{\eta}$$

$$\eta = 5, \quad \eta \rightarrow \infty$$

$$\frac{\delta^*}{\delta} = 0.34334 \quad (\eta = 5)$$

$$= 0.34415 \quad (\text{as } \eta \rightarrow \infty)$$

v_y / U AT THE EDGE of BL

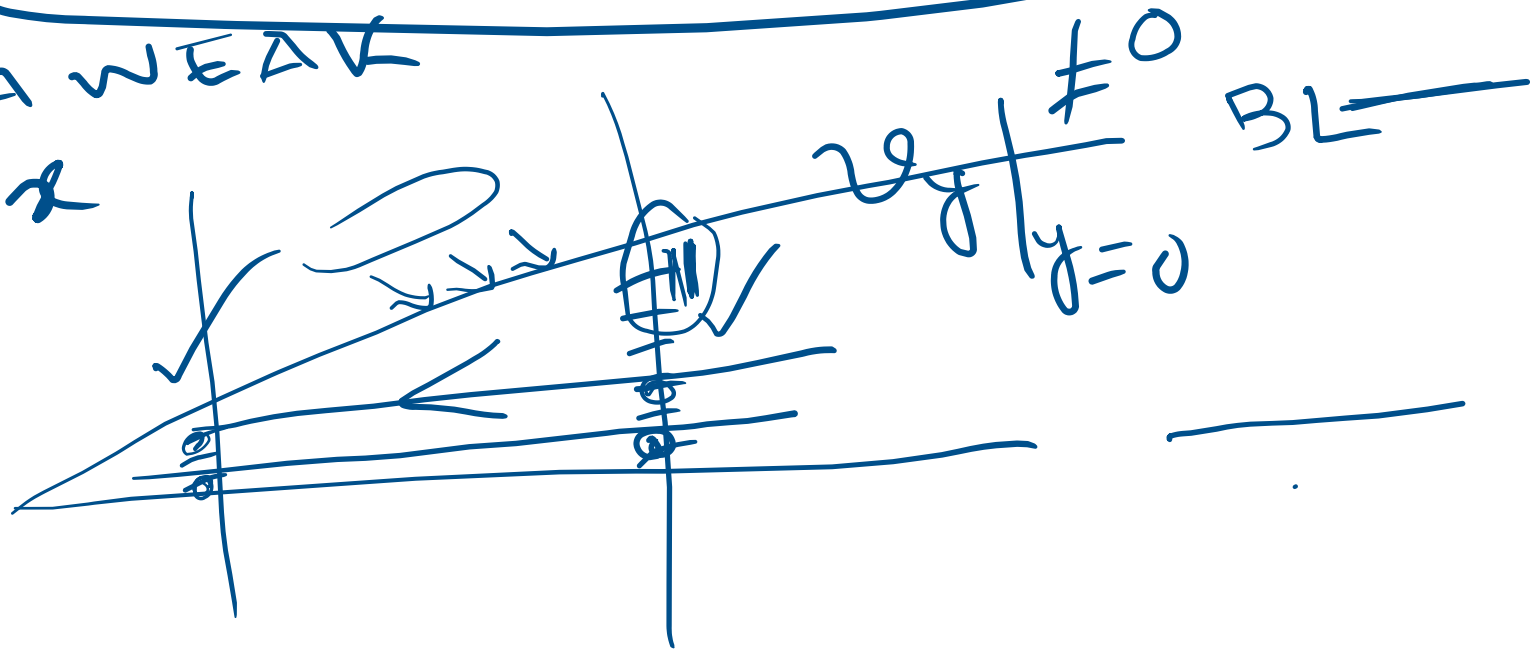
$$v_y = \frac{1}{2} \sqrt{\frac{\nu U}{x}} (\eta f' - f)$$

$$\frac{v_y}{U} = \frac{1}{2 \sqrt{Re_x}} [\eta f' - f]$$

AT THE EDGE OF BL, $\eta = 5$
 $f = 3.28$

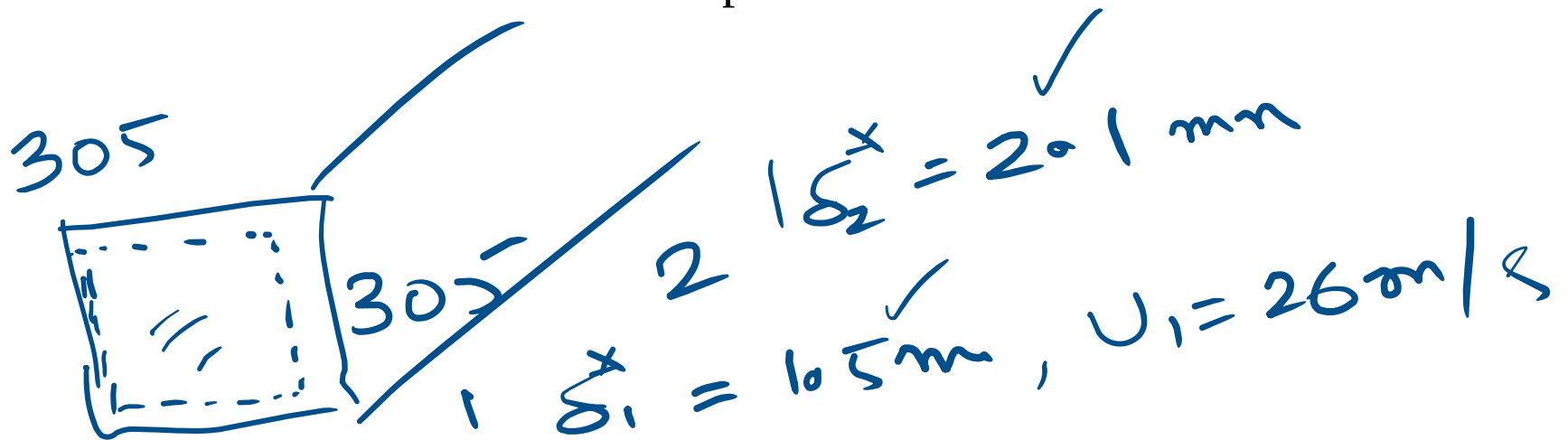
$$\frac{v_y}{U} = \frac{0.84}{\sqrt{Re_x}}$$

v_x IS A WEAK
FN of x



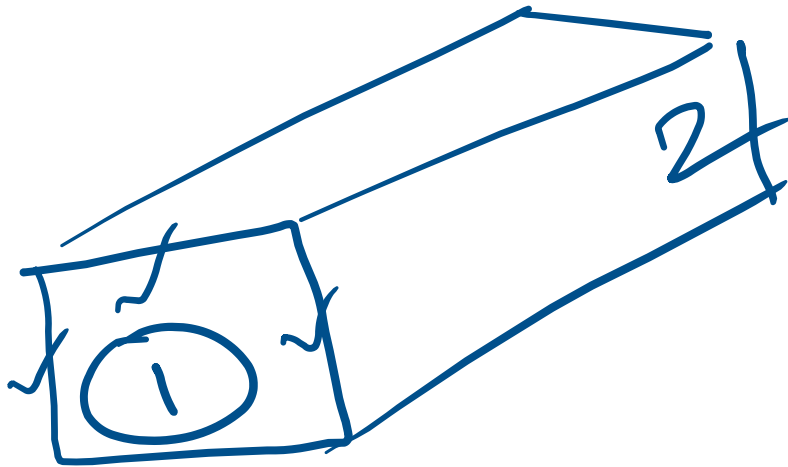
A laboratory wind tunnel has a test section that is 305 mm square. Boundary-layer velocity profiles are measured at two cross-sections and displacement thicknesses are evaluated from the measured profiles. At section 1, where the freestream speed is $U_1 = 26$ m/s, the displacement thickness is $\delta_1^* = 1.5$ mm. At section 2, located downstream from section 1, $\delta_2^* = 2.1$ mm. Calculate the change in static pressure between sections 1 and 2.

Express the result as a fraction of the freestream dynamic pressure at section 1. Assume standard atmosphere conditions.



Blasius Solution – contd.

INVISCID FLOW



$$\text{AREA} = \left(305 - \frac{2 \times 10.5}{2} \right) \times \left(305 - \frac{2 \times 10.5}{2} \right)$$

①

$$2 = \left(305 - \frac{2 \times 2.1}{2} \right) \left(305 - \frac{2 \times 2.1}{2} \right)$$

Blasius Solution – contd.

BERN. EQN

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + \cancel{gz_1} = \frac{p_2}{\rho}$$

$$+ \frac{V_2^2}{2} + \cancel{gz_2}$$

$$p_1 - p_2 = \frac{1}{2} \rho V_1^2 \left[\left(\frac{V_2}{V_1} \right)^2 - 1 \right]$$

$$\frac{p_1 - p_2}{\frac{1}{2} \rho U_1^2} = \left(\frac{U_2}{U_1} \right)^2 - 1$$

$$\frac{A_1}{A_2} = \frac{U_2}{U_1} = \frac{(L - 2\hat{\delta})^2}{(L - 2\hat{\delta}^*)^2}$$



0.016